

10(a)	speed is (directly) proportional to distance	M1
	speed is speed of recession of galaxy from an observer, and distance is the distance of the galaxy from the observer	A1
10(b)	$F = L / (4\pi d^2)$	C1
	$= (3.8 \times 10^{31}) / [4\pi \times (1.8 \times 10^{24})^2]$	A1
	$= 9.3 \times 10^{-19} \text{ W m}^{-2}$	
10(c)(i)	galaxy is moving away (from the Earth)	B1
	wavelength (of light from the galaxy) increased by the Doppler effect / due to redshift	B1
10(c)(ii)	$\Delta\lambda / \lambda = v / c$	C1
	$v = [(492 - 486) \times 3.00 \times 10^8] / 486$	
	$(v = 3.7 \times 10^6 \text{ m s}^{-1})$	
	$H_0 = v / d$	C1
	$= (3.7 \times 10^6) / (1.8 \times 10^{24})$	A1
	$= 2.1 \times 10^{-18} \text{ s}^{-1}$	